SE5 BOARD

PURPOSE

This Technical Data Sheet (for the sake of brevity, hereinafter referred to as "TDS" aims to provide an overview of the SE5 system module, outlining its main features and capabilities, and providing some technical information.
Two versions are available:

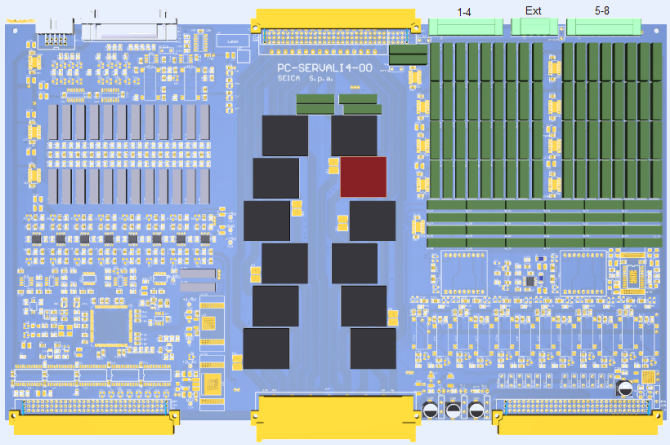
KIT PT-SERVALI5-00 for Flying Probe Line systems
KIT PT-SERVALI5-01 per Compact Line systems

SE5 MODULE DESCRIPTION

Module Description

SE5

SE5 BOARD



DESCRIPTION

The SE5 is a system module for power supply units management. It is equipped with a POWER MATRIX which can use 2 or 4 probes of the test system to supply the UUT.

TECHNICAL INFORMATION

The SE5 system module provides capabilities for:
System power supply management (up to 8 PWs units)
Use of 2 or 4 measuring probes to supply the Unit Under Test up to 2 A.
See the section "Features and capabilities" for further details
The new SE5 keeps the programming compatibility with the previous versions (SE2);
The new functions and capabilities will be managed by a new set of instructions.

ITEM P/N	APPLICATION
PT-SERVALI5-00 **PT-SERVALI5-01**	Every Seica tester is equipped with a SE5 module for power supply units management (both system and user power supply units.

SE5 BOARD FEATURES AND CAPABILITIES

The SE5 module allows to:

Manage up to 6 power supply units, implementing the following:

- voltage programming (with insulated voltage 0-5 V or 0-10 V)
- current programming (with insulated voltage 0-5 V or 0-10 V)
- Power interconnection via relay
- feedback of the output current with insulation
- feedback of the output voltage with insulation
- Sense closure via relay (to the load or to the power supply unit)

Manage 8 power supply units, implementing the following:

- voltage programming
- current programming
- programming the ON/OFF command
- feedback of the output current
- feedback of the output voltage

Management of a (2x8 PW and SENSE) matrix matrice to interconnect power probes.

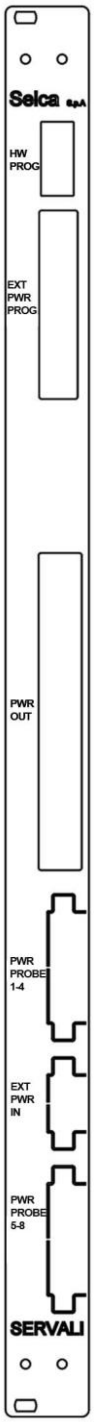
- The two power supply units can be PW2 and PW3 (ALI EXT as an alternative to PW3)
- 8 probes with PW e SENSE

HARDWARE

- Compatibility with previous versions of the SE2 modules; maintained the DIN output connector for power, the DIN 15H input power connector and the output towards the back panel of the programming signals for the six internal power supply units.
- Improved Dac and Adc (16bit) resolution for programming and sensing, unified version 60 V and 6 A old version VER 07.
- Removed external matrix for UCT1, UCT2, DMM, FNT1 instruments

CONNECTORS IDENTIFICATION AND PINOUT

The Fig. 1 shows the front plate of the SE5 in the currently available versions (KIT-PT-SERVALI5-00 and KIT-PT-SERVALI5-01)

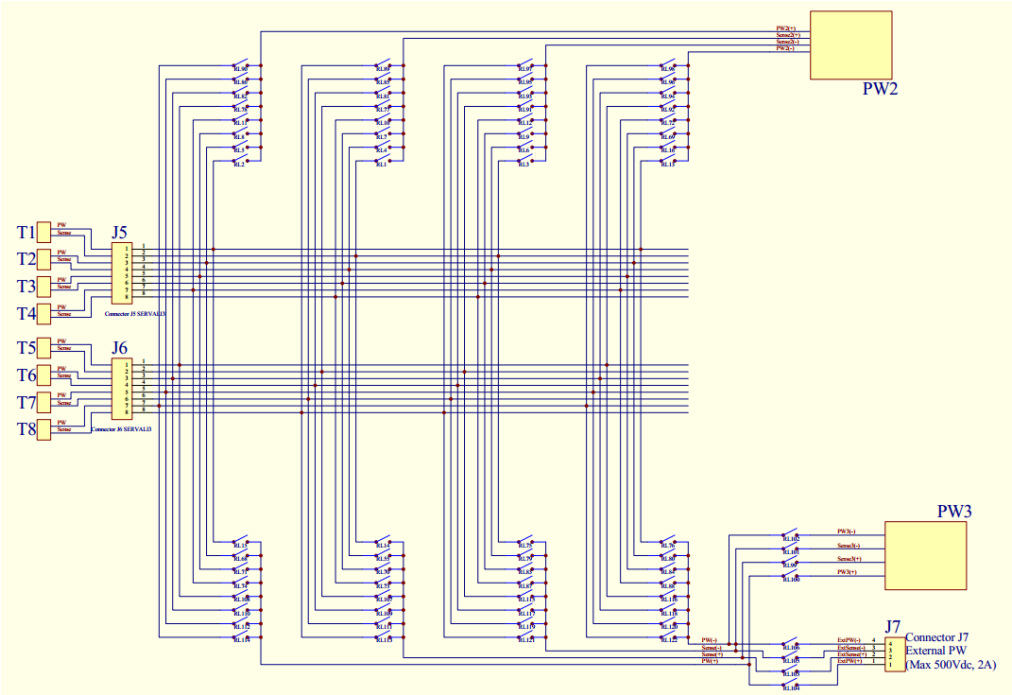


#	Connector	Silkscreen Printing
1	J6	PW RPROBE 5-8
2	J7	EXT PWR IN
3	J5	PW RPROBE 1-4
4	J3	PWR OUT
5	JP2	EXT PWR PROG
6	***J1**	**HW PROG**

NOTE: The "HW PROG" (J1) connector, is used for configurable logics programming. It is reserved to Seica technical staff.

Fig. 1: The front plate of the SE5 board

The PW RPROBE 5-8 - EXT PWR IN and PW RPROBE 1-4 connectors, are used to bring the power probe matrix on the 8 heads of the flying probers, according to the schematics below. These connectors are mounted only on the **PT-SERVALI5-00 version.**



Below the SE5 board connectors layout and pinout, and the correspondence between the output signals and column 1 of the receiver.

EXT PWR PROG (JP2) Connector			
Pin No.	**Signal**	**Pin No.**	**Signal**
1	PV_PW1	26	LV_PW1
2	PI_PW1	27	LI_PW1
3	ON_PW1	28	PV_PW2
4	GND	29	PI_PW2
5	LV_PW2	30	ON_PW2
6	LI_PW2	31	GND
7	PV_PW3	32	LV_PW3
8	PI_PW3	33	LI_PW3
9	ON_PW3	34	PV_PW4
10	GND	35	PI_PW4
11	LV_PW4	36	ON_PW4
12	LI_PW4	37	GND
13	PV_PW5	38	LV_PW5
14	PI_PW5	39	LI_PW5
15	ON_PW5	40	PV_PW6
16	GND	41	PI_PW6
17	LV_PW6	42	ON_PW6
18	LI_PW6	43	GND
19	PV_PW7	44	LV_PW7
20	PI_PW7	45	LI_PW7
21	ON_PW7	46	PV_PW8
22	GND	47	PI_PW8
23	LV_PW8	48	ON_PW8
24	LI_PW8	49	GND
25	NC	50	NC

PWR OUT (J3) Connector			
Pin	**Row C**	**Row B**	**Row A**
1	PW6+	PW6+	PW6+
2	PW6+	PW6+	SENSE6+
3	PW6-	PW6-	PW6-
4	PW6-	PW6-	SENSE6-
5	PW5+	PW5+	PW5+
6	PW5+	PW5+	SENSE5+
7	PW5-	PW5-	PW5-
8	PW5-	PW5-	SENSE5-
9	PW4+	PW4+	PW4+
10	PW4+	PW4+	SENSE4+
11	PW4-	PW4-	PW4-
12	PW4-	PW4-	SENSE4-
13	PW3+	PW3	PW3+
14	PW3+	PW3+	SENSE3+
15	PW3-	PW3-	PW3-
16	PW3-	PW3-	SENSE3-
17	PW2+	PW2+	PW2+
18	PW2+	PW2+	SENSE2+
19	PW2-	PW2-	PW2-
20	PW2-	PW2-	SENSE2-
21	PW1+	PW1+	PW1+
22	PW1+	PW1+	PW1+
23	PW1+	PW1+	SENSE1+
24	PW1-	PW1-	PW1-
25	PW1-	PW1-	PW1-
26	PW1-	PW1-	SENSE1-
27	GND_REF	GND_REF	GND_REF
28	SERV_4	GND	SERV_0
29	SERV_5	GND	SERV_1
30	SERV_6	VCC	SERV_2
31	SERV_7	+15V-	SERV_3
32	COM047	-15V	COM03

Column 1 RECEIVER	
Pin 1-32/A	**Pin 1-32/C**
PW6+	PW6+
PW6+	PW6+
PW6-	PW6-
PW6-	PW6-
PW5+	PW5+
PW5+	PW5+
PW5-	PW5-
PW5-	PW5-
PW4+	PW4+
PW4+	PW4+
PW4-	PW4-
PW4-	PW4-
PW3+	PW3+
PW3+	PW3+
PW3-	PW3-
PW3-	PW3-
PW2+	PW2+
PW2+	PW2+
PW2-	PW2-
PW2-	PW2-
SENSE4+	SENSE4-
GND_REF	GND_REF
GND_REF	PW1-
SENSE3-	SENSE3+
PW1+	PW1+
PW1+	PW1+
PW1-	PW1-
PW1-	GND_REF
SENSE2+	SENSE6-
SENSE2-	SENSE6+
SENSE1+	SENSE5+
SENSE1-	SENSE5-

PWPROBE 5-8 (J6) CONNECTOR	
Pin	**Signal**
1	PPROBE5P
2	PPROBE5S
3	PPROBE6P
4	PPROBE6S
5	PPROBE7P
6	PPROBE7S
7	PPROBE8P
8	PPROBE8S

EXT PWR IN (J7) CONNECTOR	
Pin	**Signal**
1	EXTP+
2	EXTS+
3	EXTS-
4	EXTP-

CONNETTORE PWPROBE 1-4 (J5)	
Pin	**Signal**
1	PPROBE1P
2	PPROBE1S
3	PPROBE2P
4	PPROBE2S
5	PPROBE3P
6	PPROBE3S
7	PPROBE4P
8	PPROBE4S

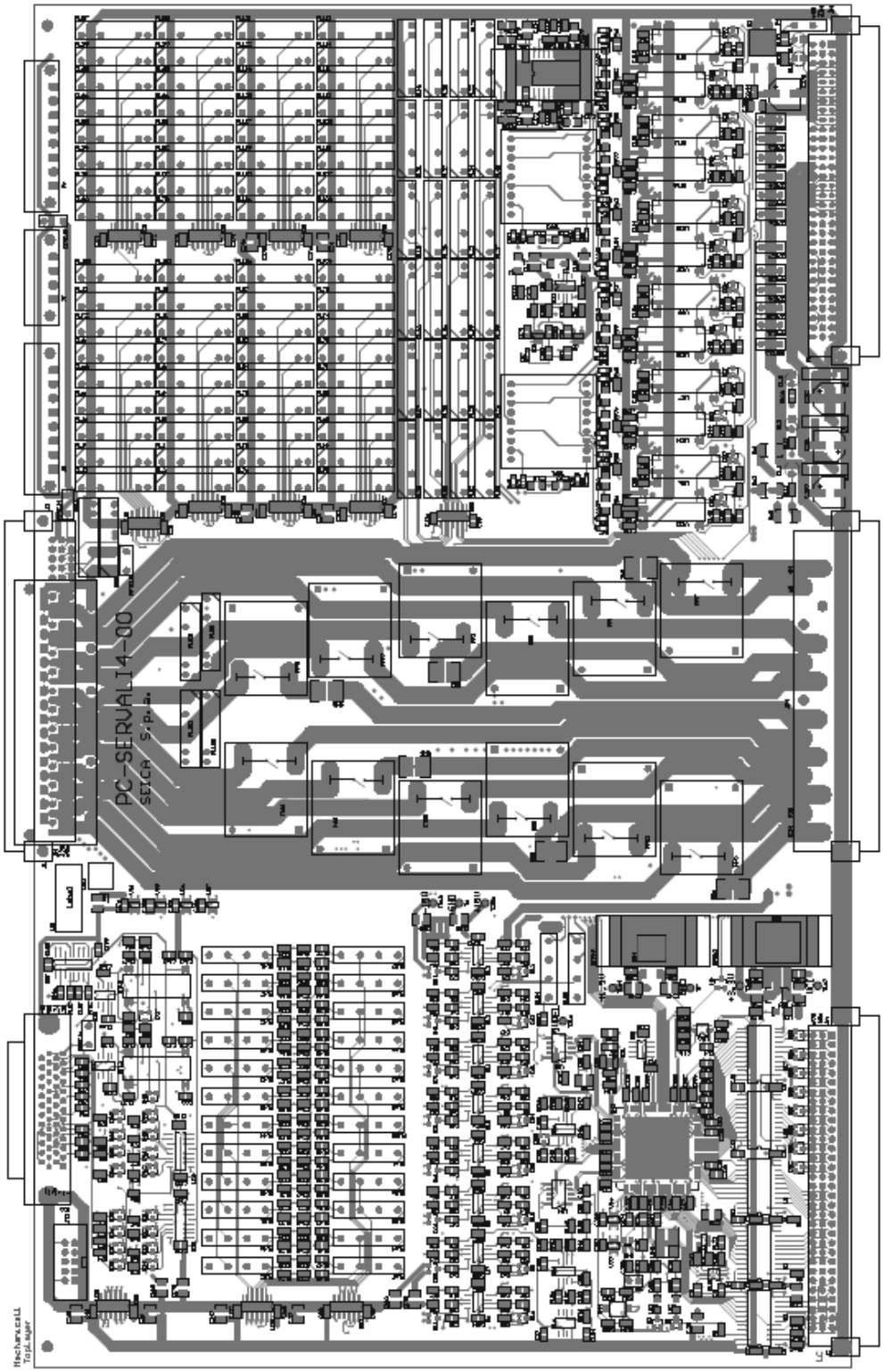
BOARD CONNECTOR TYPOLOGY

SE5	JP2	AMP M900 50-way female connector
	J1	Mini Combicon 2x16 ways female flanged (P381) connector
	J3	DIN 41612R 96-way male connector
	J5-J6	Mini Combicon 8 ways female flanged connectror
	J7	Mini combicon 4 ways female flanged (P381) connector

BOARD CONNECTORS P/N

Board	Connector	Manufacturer	ITEM P/N
SE5	JP2	FUJITSU	FCN-235D050-G/E FCN-235D050-G/H
	J3	HARTING	09031967921
	J1	PHOENIX	1843062
	J5-J6	PHOENIX	1827923
	J7	PHOENIX	1827884

SE5 BOARD LAYOUT



BOARD ADDRESS CONFIGURATION

The board addressing is not applicable for SE5 board configuration.
In the event of multiple job configuration (e.g. 2 or 4 jobs) details concerning the system resources distribution will be provided case by case.

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